Machine Learning-Based Air Quality Forecasting for the Delhi-NCR Region

A Report on Data Analysis, Model Development, and Performance Evaluation

Abstract

Air pollution in the Delhi-NCR region is a severe and recurring public health issue, driven by the overlapping effects of vehicular emissions, industrial activity, meteorological conditions, and seasonal agricultural stubble burning in neighbouring states. This report presents the data analysis and machine learning components of a web-based air quality monitoring and forecasting platform for the region, covering five NCR cities (Delhi, Faridabad, Ghaziabad, Gurugram, and Noida) over a six-year period (2020–2025).

The work combines exploratory data analysis, feature engineering, and comparative model evaluation to produce a 24–72 hour Air Quality Index (AQI) forecasting system. Three forecasting approaches were developed and benchmarked: a naive persistence baseline, a linear regression model, and a Long Short-Term Memory (LSTM) neural network operating on 7-day sequential windows of pollutant, meteorological, and fire-activity data. The LSTM model outperformed both baselines at both the 24-hour and 72-hour horizons, with its advantage widening at longer horizons — evidence that it captures genuine multi-day pollution dynamics rather than simply reproducing the previous day's reading. A lightweight full-stack dashboard was also built to serve these forecasts, though the primary contribution and focus of this report is the data analysis and modelling pipeline underlying it.

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1. Introduction

Pollution levels across Delhi-NCR fluctuate sharply with the seasons, driven in large part by post-monsoon stubble burning in Punjab and Haryana, combined with falling winter temperatures and stagnant wind conditions that trap pollutants close to the ground. Existing public information sources typically report current AQI without offering short-term forecasts or the underlying data needed to understand why levels are rising or falling.

This project addresses that gap by building a forecasting pipeline that ingests historical AQI, pollutant concentration, meteorological, and satellite fire-activity data, and produces 24, 48, and 72-hour ahead AQI forecasts per city. The emphasis of this report is on the data analysis and modelling work: understanding the structure and quality of the underlying data, engineering features that capture seasonal and short-term dynamics, and rigorously comparing forecasting approaches against honest baselines rather than reporting a single headline accuracy figure in isolation.

1.1 Objectives

- Build a clean, feature-engineered dataset suitable for time-series AQI forecasting.
- Characterise the seasonal, spatial, and cross-variable structure of pollution in Delhi-NCR through exploratory analysis.
- Develop and fairly compare multiple forecasting approaches (naive, linear, and deep learning) at 24-hour and 72-hour horizons.
- Identify and report the genuine, evidence-backed limitations of the approach rather than overstating model accuracy.
- Expose the trained forecasting model through a minimal web dashboard as a proof of concept.

2. Data and Preprocessing

2.1 Dataset Overview

The working dataset covers five NCR cities — Delhi, Faridabad, Ghaziabad, Gurugram, and Noida — with one observation per city per day from 2020 to 2025, totalling 10,960 rows. Each record includes AQI, pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO, O_3), meteorological readings (temperature, humidity, wind speed, visibility), and satellite-derived fire-activity counts intended to proxy stubble-burning intensity.

NASA FIRMS Fire Data

The fire-activity signal used in this study is derived from NASA's Fire Information for Resource Management System (FIRMS), which provides near real-time satellite-detected active fire and thermal anomaly locations from the MODIS and VIIRS instruments, commonly used as a proxy for agricultural stubble-burning activity in Punjab and Haryana. In this study, daily fire-count values for the NCR region were provided as a pre-integrated field within the working dataset rather than queried directly from the FIRMS API; each record was already aligned to the corresponding city and date prior to analysis.

2.2 Data Quality Checks

- No missing values and no duplicate (date, city) records were found across the full dataset.
- Date consistency between the date field and separate year/month/day columns was verified with zero mismatches.
- A constant 'hour' column (identical value on every row) was identified as carrying no information and was dropped, confirming the data is daily-resolution rather than hourly despite its name.
- Pollutant and fire-count columns showed 6–14% of values falling outside the interquartile range; these were retained rather than removed, since they correspond to genuine seasonal pollution spikes (e.g. the October–November stubble-burning window) rather than data entry errors.
- A correlation check found 'visibility' ($r \approx -0.96$) and 'temperature' ($r \approx -0.81$) to be implausibly strongly correlated with AQI for independently measured meteorological quantities, suggesting these fields may be partially derived from AQI in this dataset. 'Visibility' was excluded from the model feature set to avoid this data-leakage risk.

2.3 Feature Engineering

Beyond the raw pollutant and weather readings, the following engineered features were added to give the models access to short-term trend and seasonal context:

- Lag features: AQI from the previous 1, 2, and 3 days, computed per city so lags never cross city boundaries.
- Rolling statistics: 7-day rolling mean and standard deviation of AQI, and 7-day rolling mean of fire count.
- Cyclical calendar encoding: sine/cosine transforms of month and day-of-year, to represent seasonality without an artificial jump between December and January.
- A binary pollution-season flag for the October–January window.
- A binary weekend flag, to examine any traffic-related weekday/weekend effect.

2.4 Train / Validation / Test Split

Because AQI is a time series, the data was split chronologically per city (70% train, 15% validation, 15% test) rather than randomly shuffled. A random split would let the model train on days adjacent to its own test days, inflating accuracy in a way that would not hold up on genuinely unseen future data. Feature scaling (standardisation) was fit only on the training set and applied to validation and test sets, to avoid leaking test-set statistics into the training process.

3. Exploratory Data Analysis

Before modelling, the dataset was examined visually to confirm expected pollution patterns and surface any relationships worth encoding as features.

3.1 Seasonal AQI Trend

AQI across all five cities follows a strongly repeating annual cycle, with sustained severe-category readings each winter and a return to moderate/good levels each summer monsoon period. The consistency of this pattern across all six years motivated the cyclical month/day-of-year features described in Section 2.3.

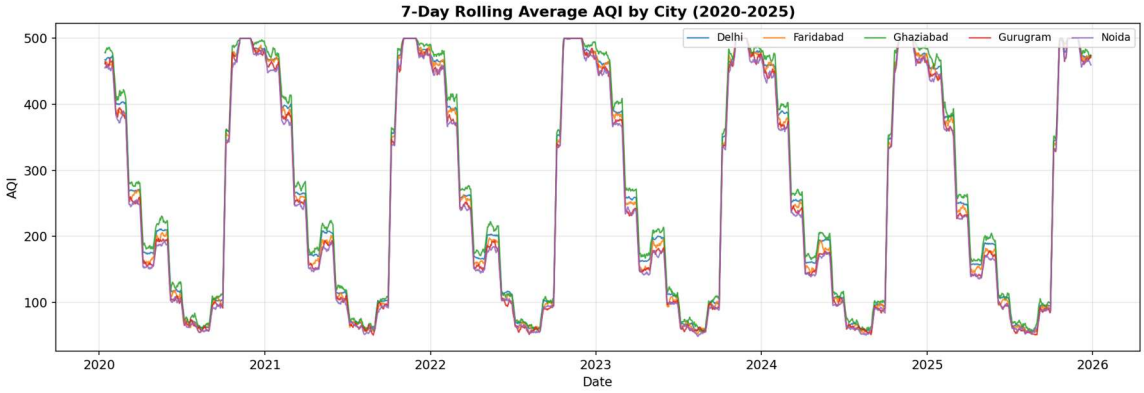


Figure 3.1: 7-day rolling average AQI by city, 2020–2025.

3.2 Monthly Seasonality

Breaking the trend down by calendar month across all years confirms that October through January consistently carries the highest and most variable AQI readings, coinciding with the stubble-burning season and winter temperature inversions, while the monsoon months (July–September) show the lowest and most stable readings.

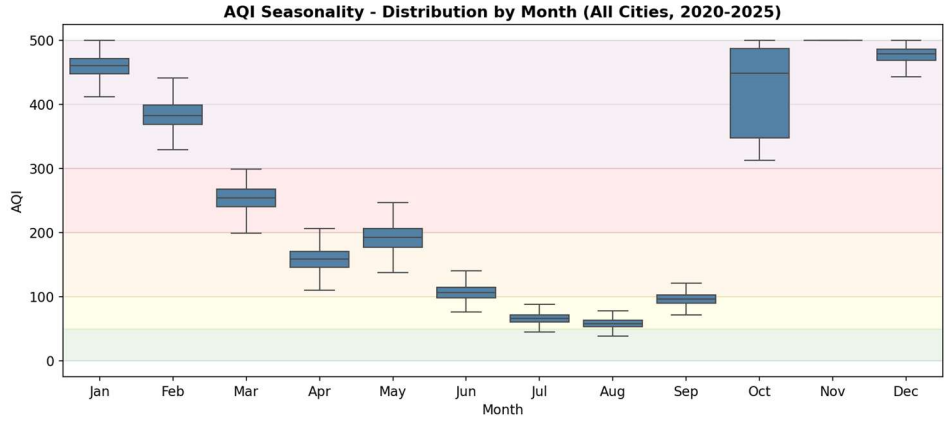


Figure 3.2: Distribution of AQI by month, all cities and years combined.

3.3 Fire Activity and AQI

Satellite-detected fire counts show a visibly stronger relationship with AQI during the October–January pollution season than during the rest of the year, supporting the use of fire count as a proxy signal for stubble-burning contribution, even though it cannot establish precise source attribution on its own.

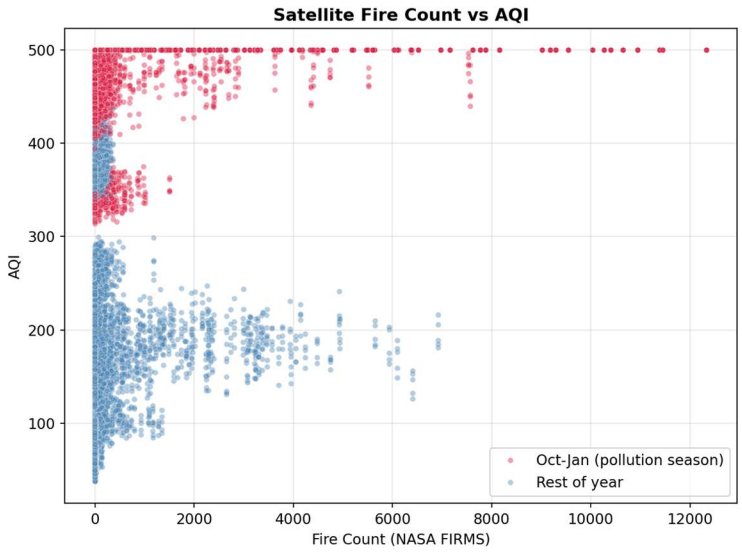


Figure 3.3: Satellite fire count vs. AQI, coloured by season.

3.4 Feature Correlation with AQI

PM2.5, PM10, CO, and NO_2 show the strongest positive correlation with AQI, consistent with their direct role in the AQI calculation itself. Wind speed and temperature are negatively correlated, reflecting how stronger winds and warmer conditions disperse pollutants. As noted in Section 2.2, visibility's near-perfect negative correlation was treated as a likely leakage signal rather than a genuine independent predictor.

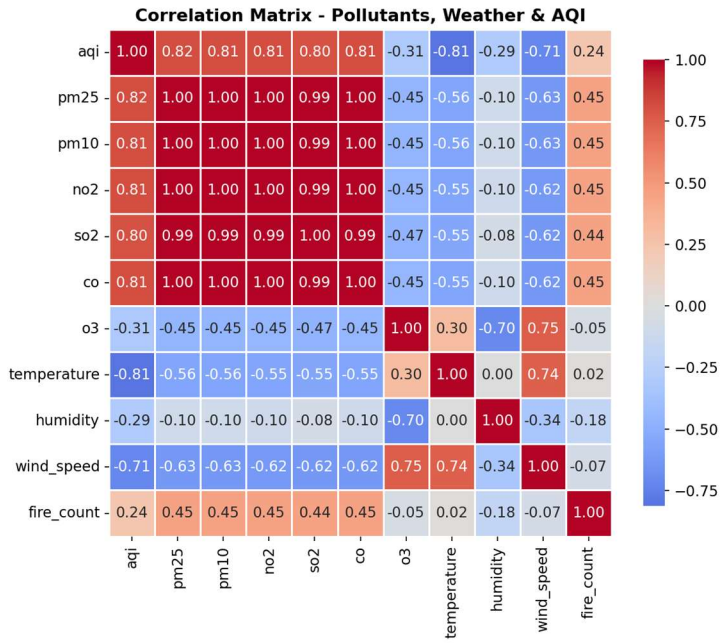


Figure 3.4: Correlation matrix of pollutants, weather variables, and AQI.

3.5 City Comparison

Average AQI is broadly similar across the five NCR cities, reflecting the fact that pollution in this region is a regional airshed phenomenon rather than one confined by city boundaries — Ghaziabad trends marginally higher, but the differences between cities are modest compared to the seasonal swing within any single city.

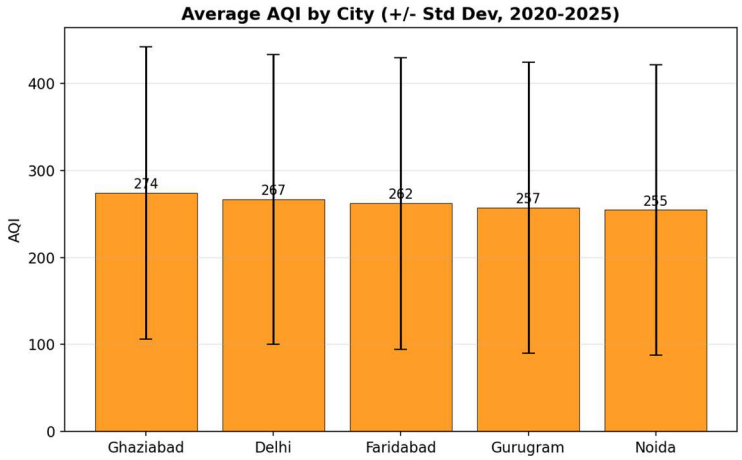


Figure 3.5: Average AQI by city with standard deviation.

3.6 Weekday vs. Weekend Effect

AQI shows only a small difference between weekdays and weekends, suggesting that at a daily resolution, traffic-driven pollution is a secondary factor compared to seasonal and meteorological drivers in this dataset.

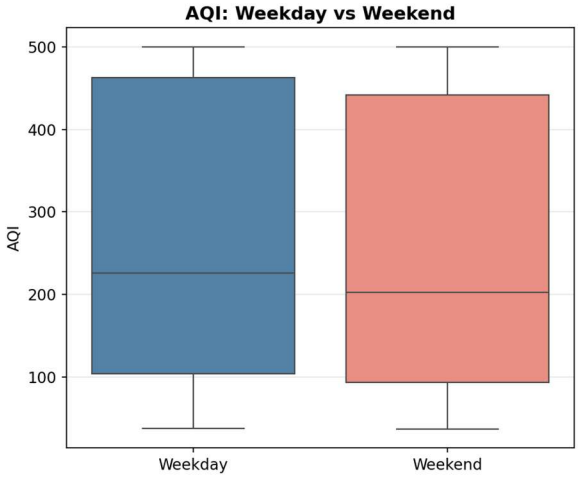


Figure 3.6: AQI distribution, weekday vs. weekend.

4. Forecasting Models

Three forecasting approaches were implemented and evaluated at two horizons (24 hours and 72 hours ahead), predicting AQI for each city from that city's most recent 7 days of history.

4.1 Naive Persistence Baseline

The simplest possible forecast — predicting that tomorrow's AQI will equal today's — was used as the reference baseline. Because AQI is strongly autocorrelated day-to-day, this baseline is deliberately difficult to beat, and is essential for judging whether a more complex model is adding genuine value.

4.2 Linear Regression

A linear regression model was trained on the full engineered feature set (pollutants, weather, lag and rolling AQI features, and cyclical calendar features) to test whether a simple linear combination of these variables could predict future AQI.

4.3 LSTM Neural Network

A Long Short-Term Memory (LSTM) network was built to exploit the sequential structure of the data directly, rather than treating each day as an independent row. The architecture consists of two stacked LSTM layers (64 and 32 units) with dropout regularisation (0.2), followed by a dense layer and a single linear output unit for the predicted AQI value. Each prediction is made from a 7-day sliding window of scaled features, built separately per city so that no sequence spans a city boundary. The model was trained with the Adam optimiser and mean squared error loss, using early stopping on validation loss to avoid overfitting.

4.4 Feature Set Note

All three models were trained on the same feature set, deliberately excluding 'visibility' for the data-leakage reasons discussed in Section 2.2, so that reported accuracy reflects genuine predictive signal rather than an artifact of the dataset's construction.

5. Results and Performance

5.1 Quantitative Comparison

Table 5.1 reports Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 for all three models at both forecast horizons, evaluated on the held-out chronological test set.

Model	Horizon	MAE	RMSE	R^2
Naive Persistence	1-Day	12.24	23.81	0.979
Linear Regression	1-Day	17.35	26.82	0.974
LSTM	1-Day	11.25	20.75	0.984
Naive Persistence	3-Day	18.97	37.39	0.949
Linear Regression	3-Day	23.01	34.71	0.956
LSTM	3-Day	14.42	26.95	0.974

Table 5.1: Model performance on the test set (lower MAE/RMSE and higher R^2 are better).

5.2 Error Comparison

Visualising MAE and RMSE side by side makes the relative gap between models clearer than the raw table alone, particularly the widening advantage of the LSTM as the forecast horizon increases.

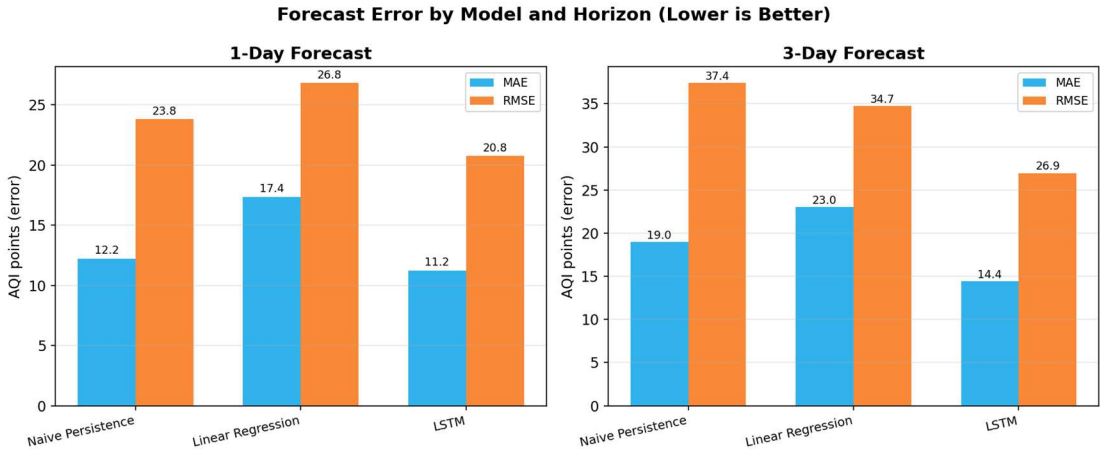


Figure 5.1: MAE and RMSE by model and forecast horizon.

5.3 Accuracy Gain Over the Naive Baseline

Because R^2 alone can be misleadingly high on autocorrelated time series (a naive "tomorrow equals today" guess already explains most of AQI's variance), the more meaningful comparison is each model's improvement in MAE relative to the naive baseline, shown below.

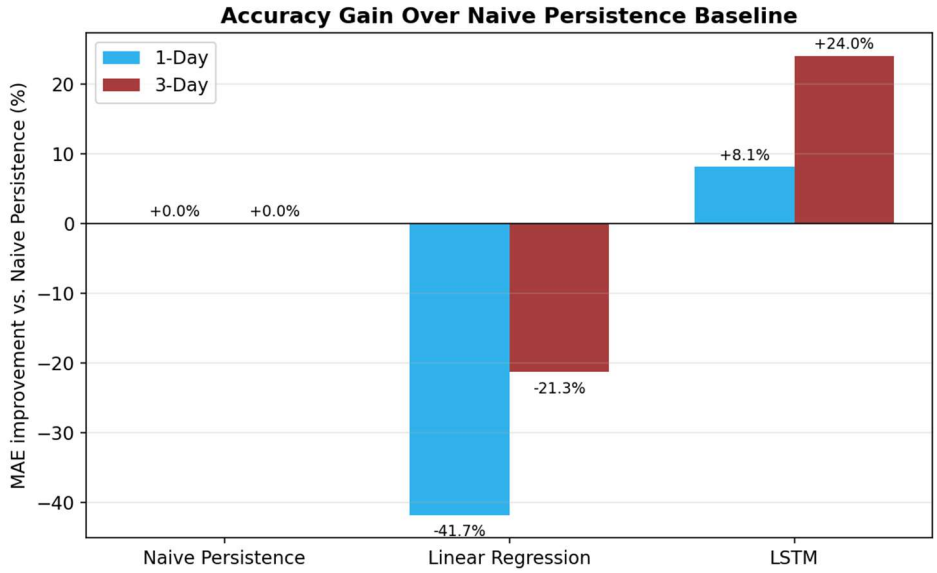


Figure 5.2: Percentage MAE improvement over naive persistence, by horizon.

5.4 Discussion

- Linear regression performed worse than the naive baseline at both horizons, indicating that AQI's relationship with the available pollutant and weather features is genuinely non-linear rather than a simple weighted sum.
- The LSTM outperformed naive persistence at both horizons, with a modest 8.1% MAE improvement at 24 hours widening to a 24.0% improvement at 72 hours.
- This widening gap is the strongest evidence that the LSTM is learning real multi-day pollution dynamics (e.g. accumulating fire activity and shifting meteorological conditions) rather than merely copying the most recent AQI reading forward, since a model that only did the latter would degrade at the same rate as the naive baseline.
- R^2 values above 0.95 across all models should be interpreted cautiously — they are substantially inflated by AQI's day-to-day persistence and the wide 37–500 range of the target variable, not solely by model skill.

6. Web Application

A minimal full-stack dashboard was built as a proof-of-concept delivery layer for the forecasting models described above. A FastAPI backend serves current AQI and 24/48/72-hour forecasts per city through a REST API, loading the trained LSTM model where available and falling back to the linear baseline otherwise. A React frontend consumes this API to display a current-AQI gauge, forecast cards with health advisories, a historical trend chart, and a city comparison view. As this report's primary focus is the data analysis and modelling work in Sections 2–5, the web layer is not elaborated on further here.

7. Limitations and Future Work

- The system currently forecasts forward from the most recent date available in the historical dataset rather than from a live data feed; integrating scheduled pulls from CPCB, OpenWeather, and NASA FIRMS APIs is required for genuinely real-time operation.
- Wind direction data was not available in the dataset, limiting the precision of any stubble-burning source-attribution claims to a correlational rather than directional analysis.
- Only 24-hour and 72-hour dedicated models were trained; the 48-hour forecast is currently interpolated between them rather than predicted by a dedicated model.
- Source attribution (estimating the relative contribution of traffic, industry, and stubble burning) remains correlational and approximate given available data, consistent with known limitations of publicly available pollution datasets.
- Future work could explore hourly-resolution data, which would give the LSTM richer sequential structure to learn from than the current daily resolution allows.

8. Conclusion

- A clean, feature-engineered, six-year daily AQI dataset for five Delhi-NCR cities was built and validated, with data-quality issues (including a likely leakage feature) identified and addressed.
- Exploratory analysis confirmed strong, consistent seasonal pollution patterns tied to the October–January stubble-burning and winter-inversion period.
- Three forecasting models were fairly compared against an honest naive baseline rather than reported in isolation.
- The LSTM model provided a genuine, widening accuracy advantage over both simpler baselines as the forecast horizon increased, supporting its use as the core forecasting engine.
- The resulting pipeline and dashboard demonstrate a practical, data-driven approach to short-term AQI forecasting, with clearly identified next steps toward a fully real-time deployment.